surface of the display to recognise fingers and objects. This technology is typically characterised by a high degree of scalability, stylus-independence, zero-force touch, high optical performance, object-size-recognition capability and low cost. But, on the other side, it is heavy, voluminous and gives a poor image quality.

PixelSense. This technology was introduced by Microsoft through a multi-touch table-top computer acting as a big banner, which incorporates hand gestures and optical recognition of objects placed on the screen. Though immune to light, it is slim and can recognise a large number of users simultaneously.

IR frame. This is a revolutionary high-performance platform, usually installed in front of any non-touch display screen to offer interactive capability. The frame is integrated with a circuit that contains a line of IR LEDs and photoreceptors. When users touch the screen, light beams are interrupted and receptors recognise the exact location of the touch. This frame is cost-effective and scalable up to 300 degrees, but it is immune to sunlight, detects even inanimate object and is difficult to clean.

IR laser pane. It is mainly used in shop windows to create a way for large interactions. It consists of an invisible laser pane that detects the reflected laser pulse of objects or users, and distance is measured from the time-of-flight of light. However, this pane has low precision without any tactile feedback.

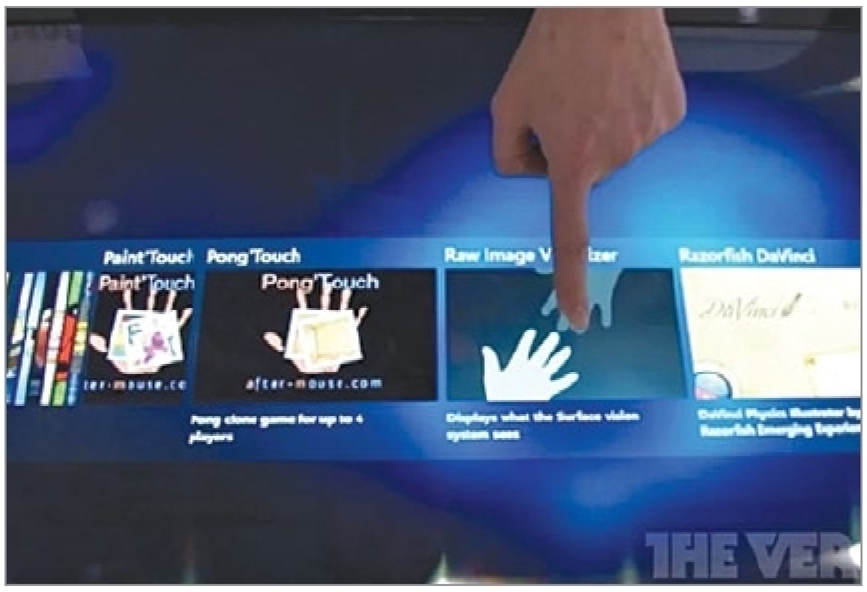

Fig. 12: IR frame (Credit: https://cdn.vox)

Fig. 13: Gaming kiosk (Credit: Kiosk industry)

Fig. 14: Innovative product presentation for the future [Credit: Eyefactive Interactive Systems (L) and Kiosk industry (R)]

InGlass. Also known as planar scatter technology, this technology uses the principle of total internal reflection that maintains light within the glass. It is mainly used in signs and billboards. Here, IR light is injected in the cover glass and extracted at the opposite side where photoreceptors placed detect the disturbance. Though the glass comes with limited touchpoints, features like invisibility, robustness due to a large number of scan lines, cost-effectiveness for large screens and highly reactive property attract consumers.

3D cameras. These use IR illumination to give depth information of the image of the object or user. The main benefit is the possibility of non-touch gesture interaction. However, this system is immune to light with no tactile feedback, and also background of the image could disturb recognition heavily.

Some examples of applications of multi-users multi-touch displays in real life are:

• Interactive multi-touch displays and touchscreens are useful in settings where more than one person operates the same digital system simultaneously. For example, gaming kiosks, which are revolutionising the gaming industry.

• In a retail store, where multi-user multi-touch displays visually stimulate shoppers to stay and explore store promotions and facilitate the shopping experience through a new engaging approach that makes customers come back.

• Interactive multi-touch switchable, smart glass screen walls and partitions used in shopping windows, bank counters, outdoor advertisements, hospitals, bathrooms and public spaces. Smart glass works on an electrical principle and switches from frosted to clear when power is applied. When power is on, the transparent glass shows a 3D stereoscopic image without affecting the display of exhibits in the shopping window.

The normal glass can be coated with a switchable optical layer using UV light, or a laminated switchable smart glass panel can be used for superior performance and clarity. Black, tinted switchable smart glass offers more privacy by turning the